Chapter 4

Results/Findings/Analysis

CSP650



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What is Result/Findings/Analysis?

- The result of research carried out to solve the problem defined
- Carries the most weight!! So spend more time writing this chapter

Analysis, Usability, Design, Implementation, Testing, Prototype or Framework



Result/Findings/Analysis

- Can be explained by phases or by objectives
- Describe how you created them, using what information, from where did you get those information, why did you decide to create them that way?
- Need to explain in detail each diagrams created



Result/Findings/Analysis

- Professional looking
- Accurate representation
- Use tables, graphs or diagrams (Labeled and titled)
- Critical analysis



Chapter	Weight	Percentage of Num of Pages	Eg (Total pages 50)
Chapter 1 – Introduction	2	12.5%	~ 5
Chapter 2 – Literature Review	3	20%	~ 15
Chapter 3 – Methodology	4	25%	~ 27
Chapter 4 – Results/Findings/ Prototype	**5**	**30%**	
Chapter 5 – Conclusion and Recommendation	2	12.5%	~ 3



Conclusion

PURPOSE

To give a summary of:

What was learned (this usually comes first)

What remains to be learned (directions for future research)

The shortcomings of what was done (evaluation)

The benefits, advantages, applications, etc. of the research (evaluation), and

Recommendations.



Sample : 01 – Requirement Result

- Interview result
- Survey result

Below the table/graph, explain output obtained

Explain every activities **findings**



CHAPTER 4

RESULT AND FINDINGS

This chapter provides the result and findings of the project. This chapter are also based on the methodology that has been discussed in Chapter 3. It consists of the achieved deliverables and discussions from each phases of the system development. The aim and objectives of the project are reflected in this chapter.

4.1 Requirement Gathering Phase

In this phase, the activity is to gather the information from the stakeholders. Conducting an interview and collect the data of staff and workflow are part of activities to get the information needed to develop Management System and Data Visualization for Security Department in Dewan Bahasa Pustaka.

4.1.1 Conducting an Interview

In order to get all the requirement from the stakeholder, interview has been performed to gather all the data. The data and requirements came from the answers that the stakeholder gave during the interview session. The answers and output to each corresponding interview questions are shown in the Table 4.1.

Table 4.1 Questions, Answers and Output from Interview Session

NO	QUESTIONS	ANSWERS	OUTPUT
1.	What is the current process of daily key request that need to be done by the staff?	Staff need to fill in key request form within the log book.	Staff ID need to be filled into the provided form. Use case identified: Manage key status

Sample : 01 – Requirement Result

- Interview result
- Survey result

Below the table/graph, explain output obtained

Explain every activities **findings**

How it related with your use case

Table 4.1 Questions, Answers and Output from Interview Session (Continued)

2.	How does staff can issue a complaint towards security department if they face any security-related problem?	Staff need to download the complaint from DBP office virtual system and may hand it manually towards security department or through email.	Staff complaint are generated Use case identified: Create complaint and update complaint
3.	What is the process for staff in order for them to request for vehicle park for several days?	Staff need to download the complaint from DBP office virtual system and fill in vehicle details then send it to security room.	Use case identified: Request parking space and update parking request.
4.	How can staff obtain the keys during the weekend? Does it differ compared with the weekday process?	Staff need to fill in key request form which can be downloaded from the same virtual system. However, the staff need to obtain approval from the head of department.	Staff need to fill information and obtain approval from head of the department. Use case identified: Request key and update key request.

Based on the analysis shown in Table 4.1, the procedures workflow and attributes of staff were identified. The questions for the processes of each application form were also identified. The last question is different which is they need to acquire signature from the head of department to take keys during weekend or public holiday. The answer from question 1 and question 2 was analysed and the stakeholder give the workflow of current manual procedures. Based on the observation, the flow of the system was identified by having this workflow. It is important to fully understand the flow of the current manual procedures so that the flow of the system can be determined.



Sample : 01 – Requirement Result

- Interview result
- Survey result

Below the table/graph, explain output obtained

Explain every activities **findings**

Every figure and graph must come with a paragraph to explain the findings

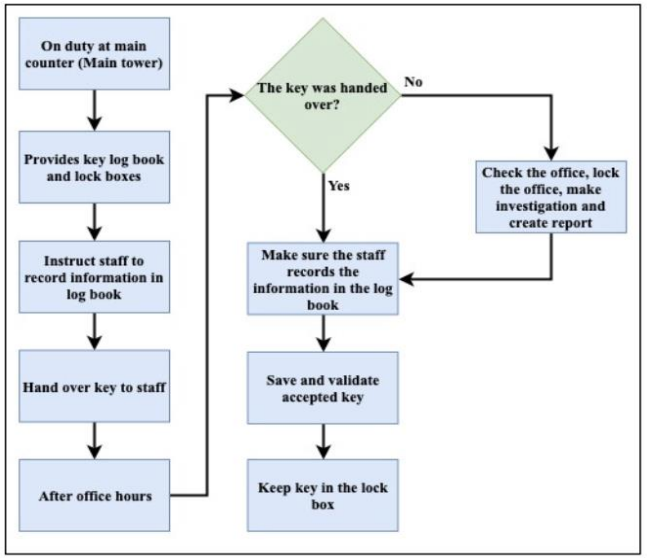


Figure 4.1 Flowchart current procedure to take key

From the workflow shown in Figure 4.1, the flow shows the current procedure to take key over the counter. The staff in charge at the main counter checks the register key and lock boxes for key. After that, staff need to fill their information in the log book including date and time for take that key and staff in charge hands over key to staff. After office hours, the key need to be handed over to the main counter. The staff that handed over the key need to fill the information into the log book. If the key is not handed over, staff in charge need to check the office and lock the office if there is no one within the office anymore. Security department conducts an investigation and lodge a report for the unreturned key. The staff in charge also need to validate accepted key and keep the key in the locked box.



Sample : 02 – Analysis Result

- Analysis from interview – obtained use cases
- Explain how every use solve which problem

Explain every activities **findings**

Every figure and graph must come with a paragraph to explain the findings



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4.2.1 Use Case Diagram

Based on the Figure 4.3, there are 11 use cases that satisfies the need of the stakeholder and the user for this system. There are 3 actors for this system which are Admin of the Department and the Staff Department that administrate the system, while the Staff are the user for this system. The Admin Department able to create the Security Staff Department, which then lead the security department to create another Staff Department. Staff in charge on main counter able to handle the key flow from security room. Staff also can make complaint, request parking for their vehicle and request to take the keys during weekend.

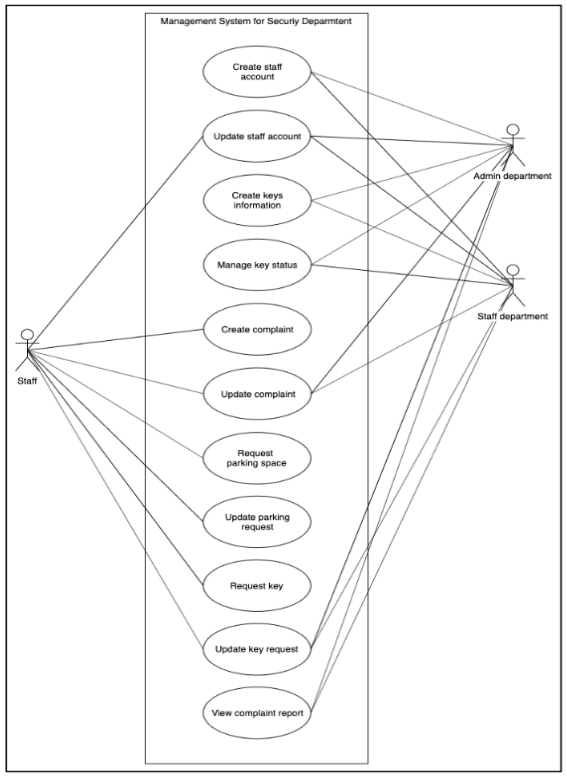


Figure 4.3 Use Case Diagram for MSSD

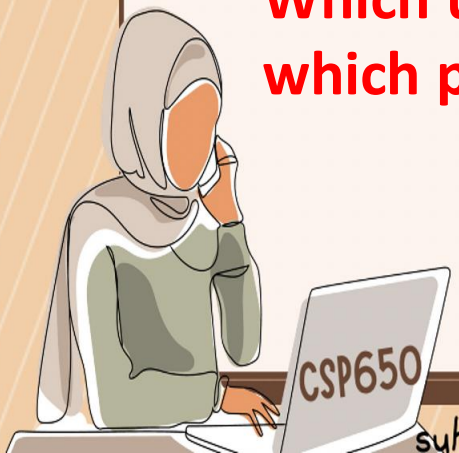
Sample : 02 – Analysis Result

- Analysis from interview – obtained use cases
- Explain how every use solve which problem

Explain every activities **findings**

Every use cases come with an explanation.

Which use case solve which problem



Create staff account

Only admin can create staff department and only staff department can create other department.

Update staff account

Staff can update account details such as name, position, level, department and phone number.

Create keys information

Staff in department can create key information such as department, level and total key.

Manage key status

Staff in department can manage key by taking staff ID and verify them by their department and the key department that attach on the key.

Create complaint

Staff can create complaint and directly submitted to security department.

Update complaint

Staff can update complaint and staff in security department can give feedback to staff about progress on the complaint.

Request parking space

Staff can request to parking their vehicle for several days by entering their vehicle information such as plate, type, model, color and parking level.

Update parking request

Staff can update their parking vehicle information.

Request key

Staff can request to take keys during weekend by entering the information such as staff details, reason and date requested.

Sample : 02 – Analysis Result

- Analysis from interview – obtained DCD
- Explain relationship among all entities

Explain every entities **findings**

Every entity come with an explanation.

Which relationship indicates which use cases



4.2.3 Domain Class Diagram

The Figure 4.5 shows the domain class for the Management System for Security Department. The domain class consists of eight (8) tables which are the RequestKeyWeekend, Complaint, Keys, Vehicle, Staff, KeysRegistration, SecurityDepartment and OtherDepartment. Based on the flow of the key in and out key from security room, KeyRegistration table are used to save the data about staff and key. The complaint details are stored within the Complaint table, as well as the Vehicle and the RequestKeyWeekend table that store their own details as the attributes. SecurityDepartment and OtherDepartment are using the details from Staff table.

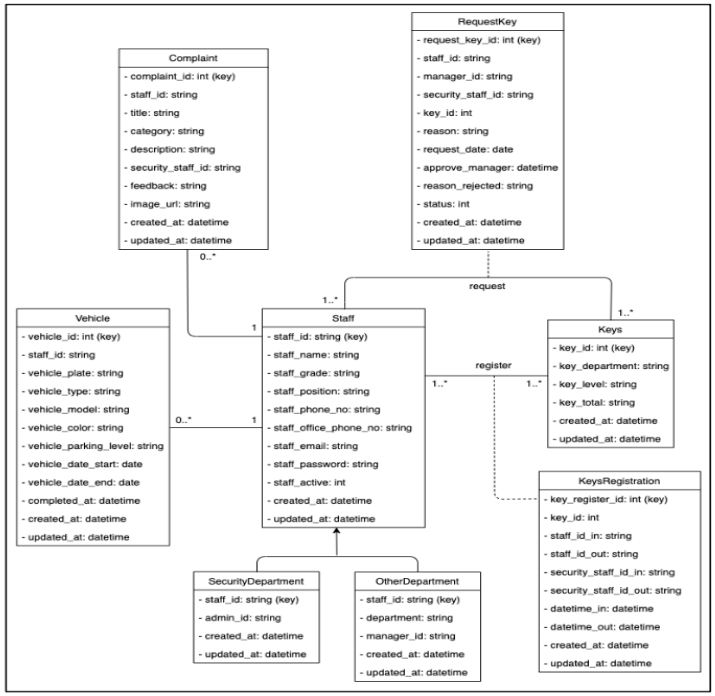


Figure 4.5 Domain Class Diagram

Sample : 03 – Design result

Results of design:

1. **Storyboard**
2. MLSD
3. ERD design

4.3.1 Storyboard of The User Interface

Figure 4.7 show the storyboard for the staff security guard to record flow keys in and out from security room. This is where the security guard need to fill the form such as staff ID, department and level. After submitting the data, it then saved into particular table on the date performed.

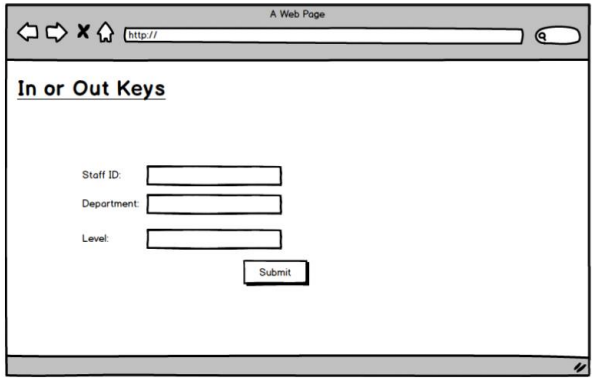


Figure 4.7 Storyboard for manage key status

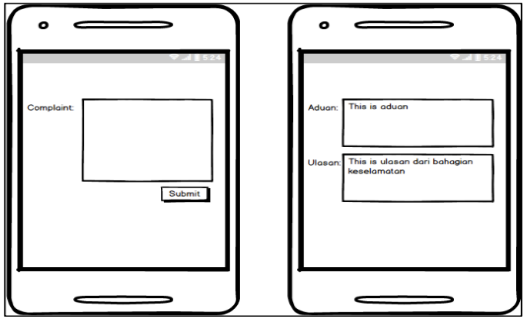


Figure 4.10 Storyboard for complaint form

Figure 4.11 shows the page for the request to park their vehicle. Firstly, the staff need to select their registered vehicle. After submitting the form, staff can view the current vehicle that are being park at the basement.

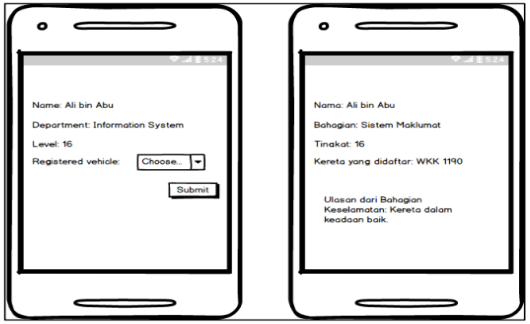
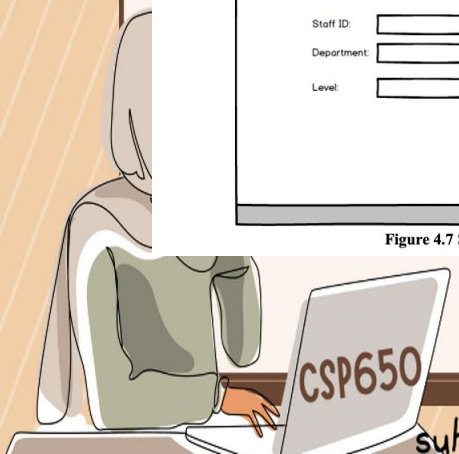


Figure 4.11 Storyboard for parking vehicle

Figure 4.12 shows the form of request to take keys during weekend. The first step is staff need to fill the form by selecting the date that they need to take the keys. After submitting the form, the staff need to wait for approval from the head of department in order to take the key on the date.



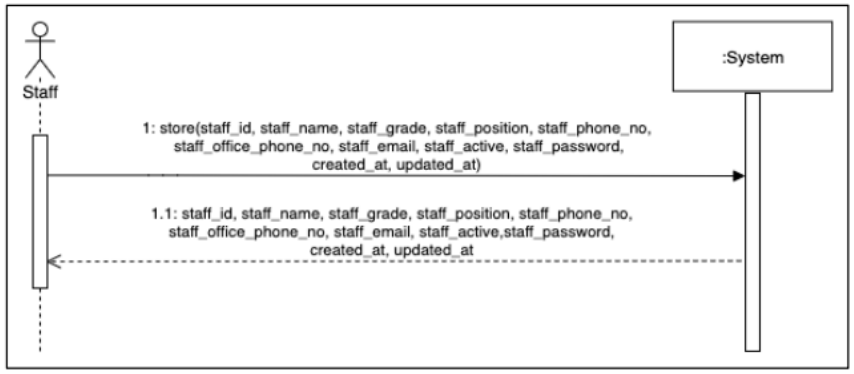
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Sample : 03 – Design result

Results of design:

1. Storyboard
2. **MLSD**
3. ERD design

System Sequence Diagram (SSD) were designed to show the complete interaction between the actor and the system. Figure below shows the system sequence diagram for the register staff department.





Sample : 03 – Design result

Results of design:

1. Storyboard
2. MLSD
3. **ERD design**



4.3.2 Entity-Relationship Diagram

The Entity Relationship Diagram (ERD) for Management System and Data Visualization for Security Department in DBP consist of 8 table which are RequestKeyWeekend, Complaint, Vehicle, Keys, KeyRegistration, Staff, SecurityDepartment and OtherDepartment. The Figure 4.14 shows all the primary keys and foreign keys with the related tables.

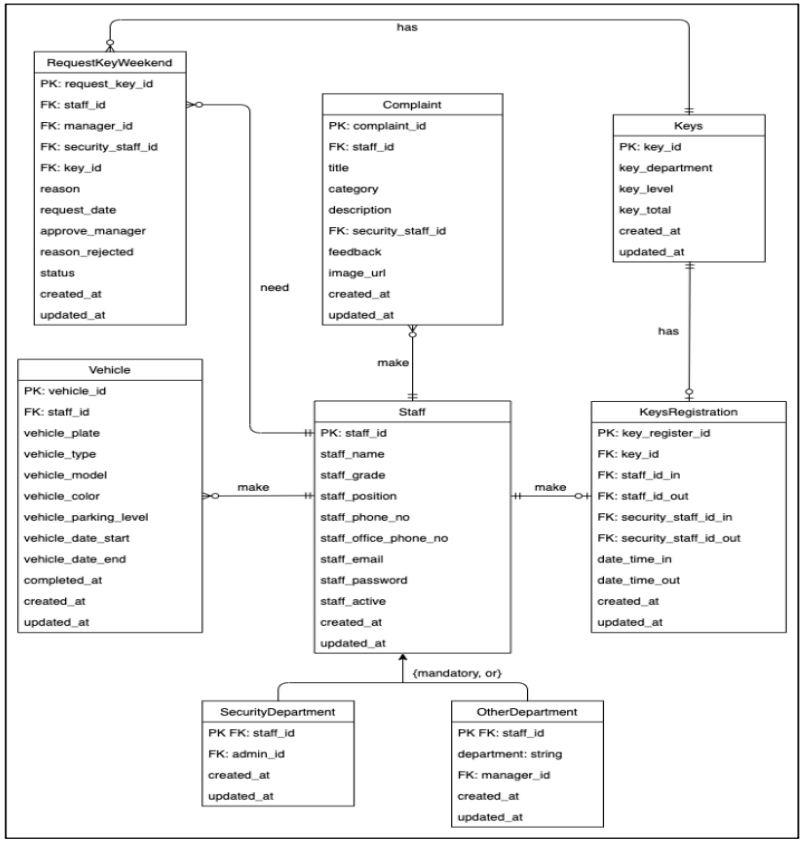


Figure 4.14 Entity-Relationship Diagram

Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case by use case

Explain every button and menu

a) Create Staff Account

Figure 4.17 shows the interface of add staff for Security Department. This page only can be accessed by admin in the Security Department. In order to register the staff, admin need to fill all the required information which then saved into the database.

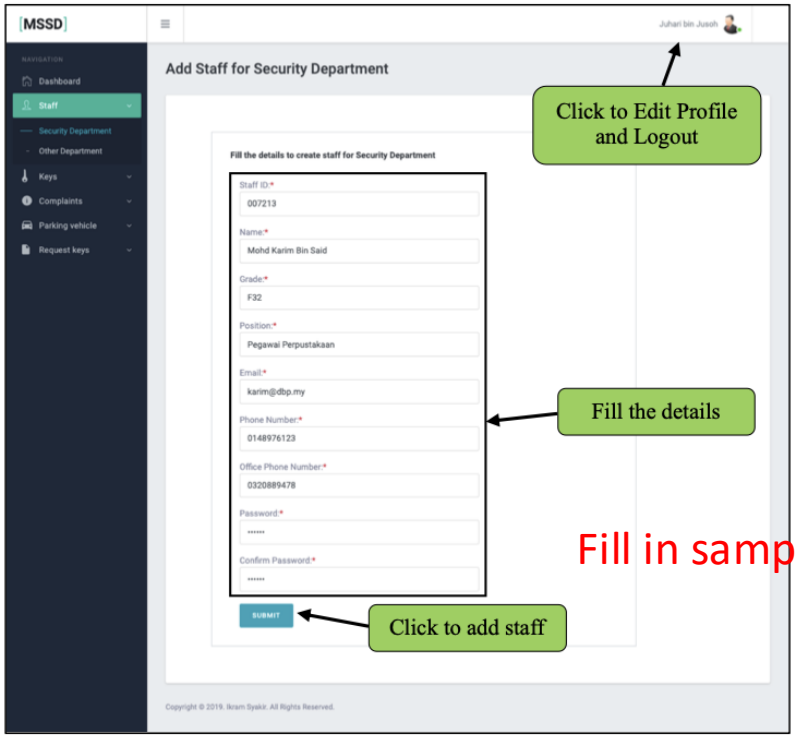


Figure 4.17 Create staff security department

Fill in sample data



Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case by use case

To show update use case, screenshot data before and after updating

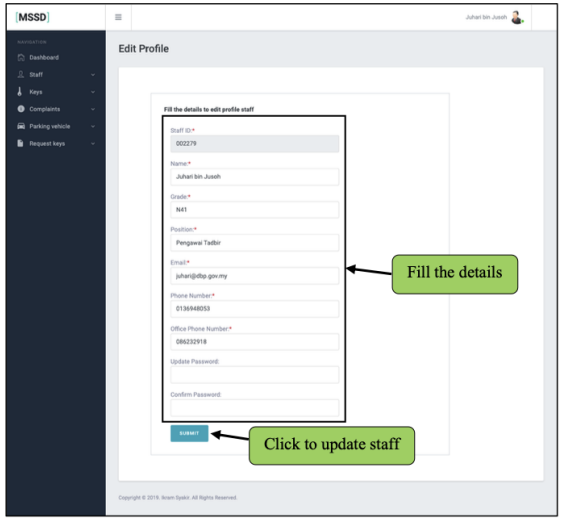


Figure 4.19 Update staff security department

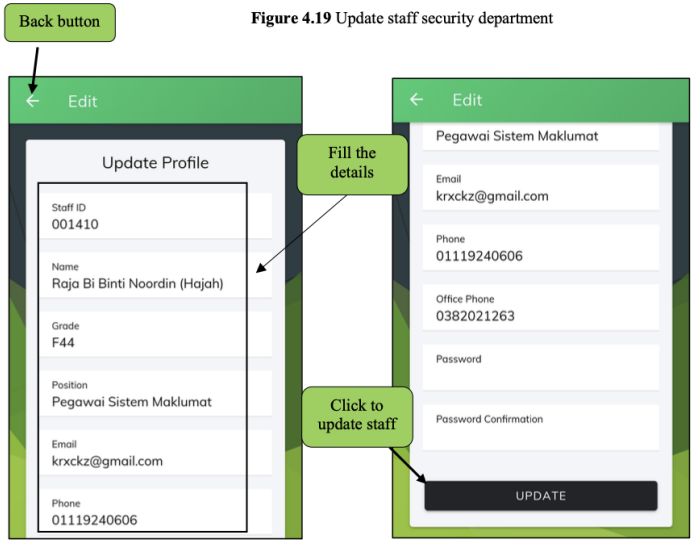


Figure 4.20 Update staff other department



Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case by use case

Explain function for every use case

Elaborate the use case is solving which problem



c) Create Key Information

Figure 4.21 shows the interface of creating key. The admin and staff can create key through the form provided in the system. The key information then be saved in the database which can be used to record flow in and out key from security room.

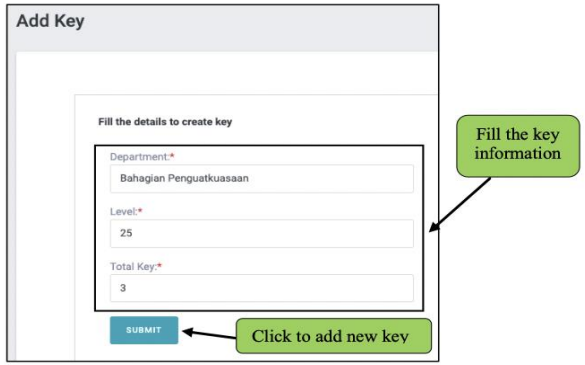


Figure 4.21 Add key information

d) Manage Key Status

For the registration key, there are two option which are in and out. Figure 4.22 shows the interface for the key in and out. Each registration key is registered on that date.

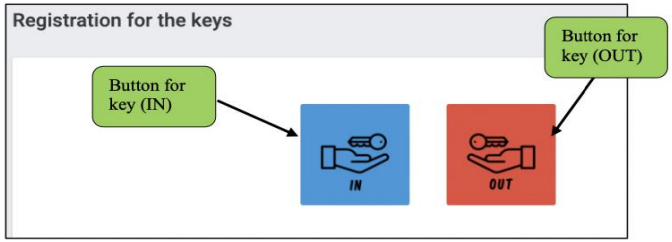


Figure 4.22 Key registration process

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Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case by use case

If one use case have many pages of systems, explain step by step the process with sample data.



Figure 4.23 shows the interface of registration key for in of the day. The staff security guard search by entering the staff ID for that staff. After that, user then proceeded to the confirmation page.

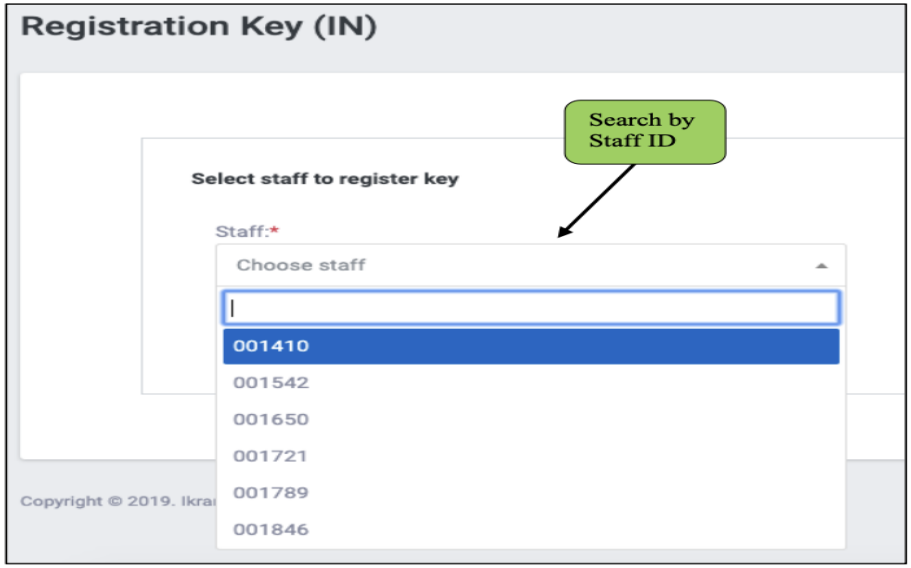


Figure 4.23 Select staff to register key



Figure 4.23 Select staff to register key (Continued)

Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case by use case

If the result of use case/pages involves images, please use REAL images with suitable data. Not dummy images.

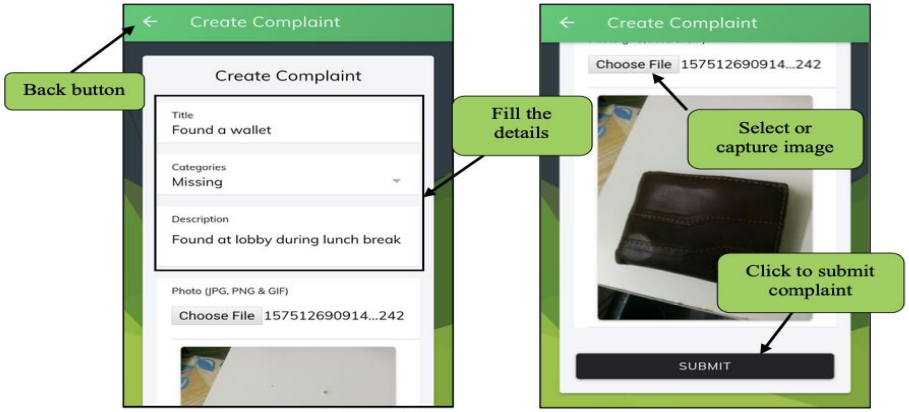


Figure 4.29 Create complaint

f) Update Complaint

Figure 4.30 shows the interface of update the complaint. The staff can update the information about the complaint and also can retake the picture or change the picture. The complaint cannot be updated if the staff from Security Department already give feedback about the complaint.

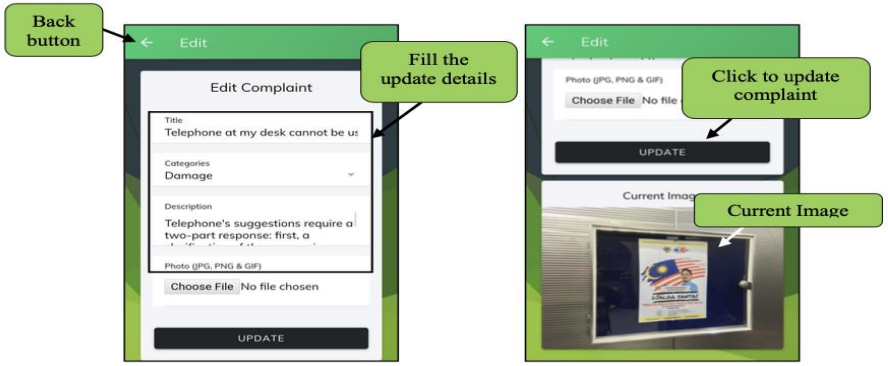


Figure 4.30 Update complaint



Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case
   by use case

SHOW clearly the result of your additional features:

Example

1. Email notification
2. SMS notification
3. Prediction graph/result
4. Geolocation map
5. Tracking monitor visual



Figure 4.36 shows the process of approving the requested key from the staff. The head of department only sees this page and can approve or reject the request by clicking button "Approved" on the application. After that, a popup showed up to confirm the request. The web application automatically sends an email to the staff about the request that has been approved by the head of department.

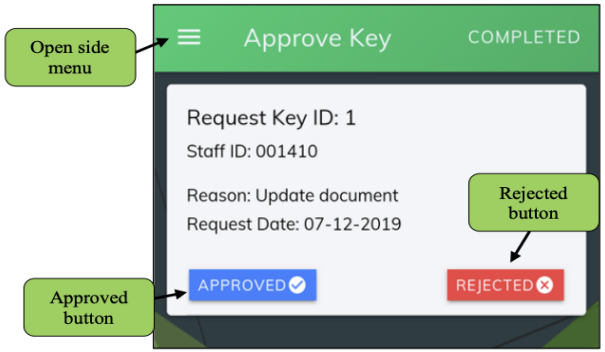
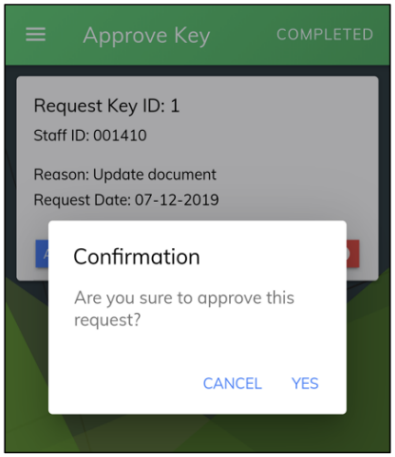


Figure 4.36 Approve request key



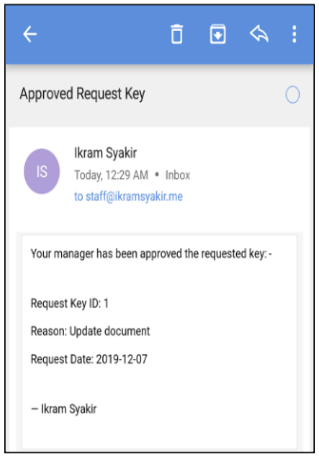


Figure 4.36 Approve request key (Continued)

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Sample : 04 – Implementation result

Results of systems:

1. Screenshot use case
   by use case

Visualize your findings/ usability testing

Example

1. graph
2. Pie chart
3. Prediction graph/result
4. Geolocation map
5. Tracking monitor visual



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4.4.3 Data Visualization

In this section shows the suitable chart for data visualization which is using Bar Chart. Figure below shows the visualization chart that using library from Google Chart. Based on the Figure 4.38, it shows the last month of each category for the total complaint.



Figure 4.38 Bar chart visualization of total complaint for last month

Figure 4.39 shows selected date for data visualization of total complaint. The staff need to select the date to show the data. After submitting the selected date, the visualization chart are shows based on the selected date.

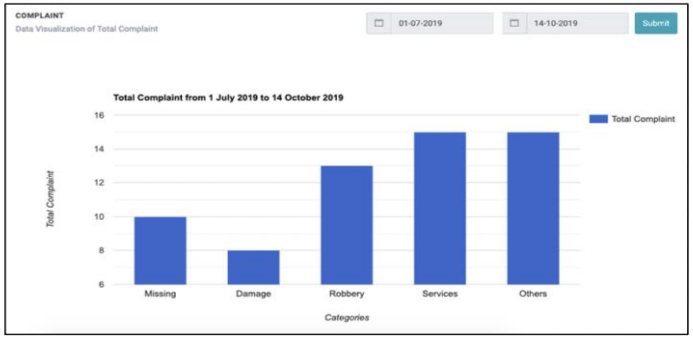


Figure 4.39 Selected date for total complaint

Result & Findings

- **PURPOSE**
  - to present the results and make them meaningful to the reader.
- **WHAT THE RESULT SECTION INCLUDES**
  - **statement of results: the results are presented in a format that is accessible to the reader (e.g. in a graph, table, diagram or written text). Notice that raw data is usually put in an appendix, if it is included at all.**
  - explanatory text: all graphs, tables, diagrams and figures should be accompanied by text that guides the reader's attention to significant results. The text makes the results meaningful by pointing out the most important results, simplifying the results (e.g. "nearly half" instead of "48.9%"), highlighting significant trends or relationships.



Result & Findings

- **COMMON PROBLEM**
  - the text includes too much detail that simply repeats data presented in graphs, tables, etc. without making the results meaningful.
  - Solution: remember that tables etc. are used to present a lot of information efficiently, but that your job is to direct the reader's attention to significant parts of this information.



Result & Findings Analysis

ORGANIZATION

There are two basic ways of organizing the results:

- Presenting all the results, then giving a discussion (perhaps in a different section)
- Presenting part of the results then giving a discussion, presenting another part then giving a discussion, etc.

- The method of organization you use will depend on the quantity and type of results you obtain from your PROJECT/research. You should look for a method of presentation that makes the information and ideas you are presenting as clear as possible to the reader.



Result & Findings Analysis

Results

Figure 2 shows that the most frequent diagnosis strategies were "Historical information" (29% of the 182 observed strategies), "Least effort" (11.5%), "Reconstruction" (9.8%) and "Sensory check" (8.7%) (see Appendix). Strategies such as "Historical information", which check available information about the failure history, and "Least effort" are two low cost technical checks which shorten the time needed for diagnosis activities (see Appendix). Strategies such as "Split half", leading to a binary reduction of the problem space, and "Information uncertainty" play only a minor role in real-life failure diagnosis of machine tools (1.1%, see Figure 2).

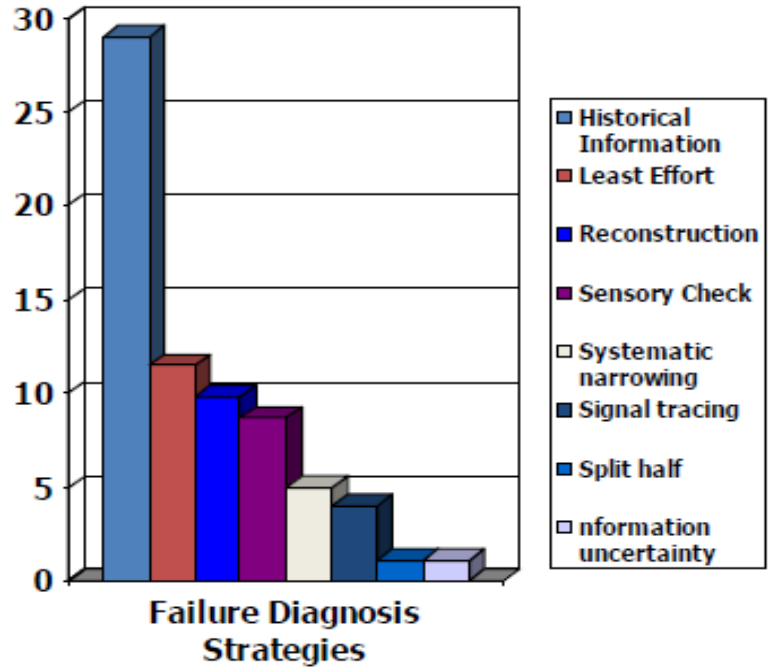


Figure 2. Frequencies of Failure diagnosis strategies



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Result & Findings Analysis

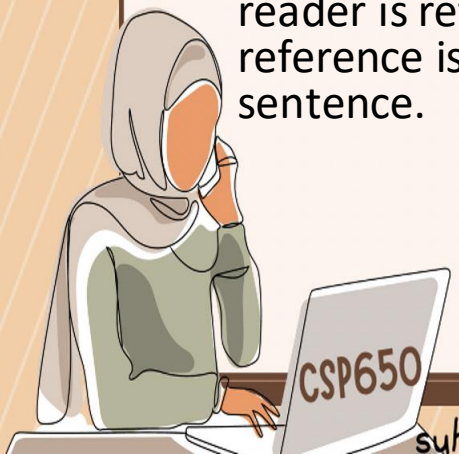
Results – Text analysis

"**Figure 2 shows**"

Reference to a figure. Notice that the present tense is used ("shows"). Usually references to figures, tables etc. are put in parenthesis rather than in the main body of the sentence because they are of secondary importance to the results themselves.

"**the most frequent diagnosis strategies were**"

The writer is pointing out the significance of three of the results (i.e. that they were the most frequent diagnosis strategies). Notice that the writers refers to the figure containing the information ("Figure 2 shows"), and that detailed percentages are de-emphasized by being included only in parentheses. In fact, this detailed information does not need to be included in the text since it appears in the figure. For more detailed information, the reader is referred to the appendix: "(see Appendix)." Notice that this reference is in parentheses too as it is not part of the main body of the sentence.



Result & Findings Analysis

Results – Text analysis (cont'd)

"are two low-cost technical checks which"

The writer is summarizing the benefits of two of the strategies in order to indicate why they were most frequently used. Although this is in fact discussion, it is helpful for the reader to have this information while looking at the results. A detailed discussion of the results appears in the Discussion section of the same article.

"play only a minor role"

The writer is pointing out the least frequent strategies. In this case, the low frequency of these strategies is of interest (see the Discussion section of the same article) and therefore the reader's attention is directed towards them.



Result & Findings Analysis

ADVICE

References to the figure and to the appendix are generally put in parentheses, e.g. "(see Appendix)" because this information is of secondary importance. Of primary importance are the results themselves, so most of the sentences focuses on them. Look at the following two sentences. Which one is **more effective**? Remember the purpose of the text in a results section.

a) Table 1 shows the results from the laboratory experiment.

b) The results from the laboratory experiment indicate that the reaction **proceeds faster** in the presence of this metal (see Table 1).



Result & Findings Analysis

"**a) Table 1 shows the results from the laboratory experiment**."

The point of this sentence is just to tell the reader to look at Table 1. It does not make the results in Table 1 meaningful because it does not comment on them.

"**b) The results from the laboratory experiment indicate that the reaction proceeds faster in the presence of this metal (see Table 1)**". This sentence is more effective than a) because it makes the results in the table meaningful by pointing out a relationship between the speed of the reaction and the presence of the metal. Notice that the reference to the table is de-emphasized by being put in parentheses because it is of only secondary importance.



Result & Findings Analysis

PURPOSE AND CONTENT OF THE DISCUSSION SECTION

The discussion section is for comment on and explanation of the results.

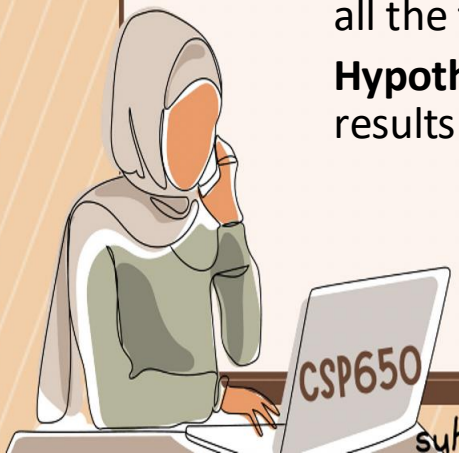
It includes:

Explanation of results: the writer comments on whether or not the results were expected, and presents explanations for the results, particularly for those that are unexpected or unsatisfactory.

References to previous research: comparison of the results with those reported in the literature, or use of the literature to support a claim, hypothesis or deduction.

Deduction: a claim for how the results can be applied more generally (a conclusion based on reasoning from the results, e.g. we fed fish a new feed, all the fish gained weight, therefore the new feed causes fish to gain weight).

Hypothesis: a more general claim or possible conclusion arising from the results (which will be proved or disproved in later research).



Result & Findings Analysis

COMMON PROBLEM

The discussion does not discuss - simply supplies more detail about the results obtained.

Solution: remember that the discussion should **explain** the results.



Result & Findings Analysis

ORGANIZATION

There are two basic ways of organizing the results and discussion:

1. Presenting all the results, then giving a discussion (perhaps in a different section)
2. Presenting part of the results then giving a discussion, presenting another part then giving a discussion, etc.

The method of organization you use will depend on the quantity and type of results you obtain from your project/research. You should look for a method of presentation that makes the information and ideas you are presenting as clear as possible to the reader.



Result & Findings Analysis

Refer to the article in the Result Section

Discussion

These strategies appeared in about 60% of the total observed strategies. The primary strategy was “Historical Information”. This corresponds to the results of Hoc (1989). In information theory, strategies such as “Information uncertainty”, which eliminates the greatest number of failure causes, or “Split half”, which results in a binary splitting of the problem space, are economical ways to shorten the problem space. We found that in real-life failure diagnosis, even maintenance experts with more than 20 years experience seldom used these strategies. One reason may be that the use of this strategy requires information about conditional probabilities and a fully described problem space that cannot be supposed for troubleshooting in complex manufacturing systems.



Result & Findings Analysis

Discussion - Text Analysis

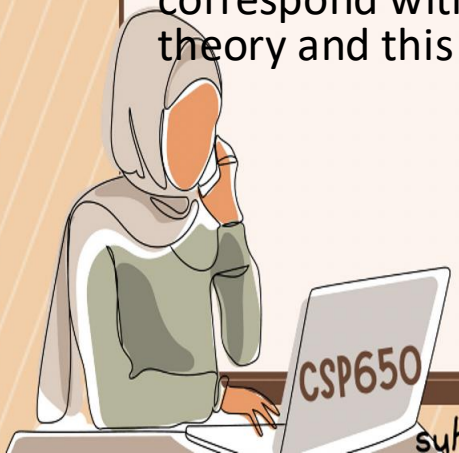
"The primary strategy was “Historical Information" : Up to this point, the writer is providing a summary of the results by highlighting the most significant findings.

"corresponds to the results of Hoc (1989)“: The writer supports the validity of the results by referring to similar results obtained by another researcher.

"In information theory“: The writer summarizes relevant information from the theory. Notice how this is important to put the writer's own research into context (see the next sentence).

"We found that“: The writer summarizes the results in order to show that they differ from what the theory would predict (the theory was summarized in the previous sentence).

"One reason may be“: The writer suggests why the results from this study do not correspond with the theory, i.e. the writer is explaining the difference between theory and this study's results outlined in the previous sentences



Result & Findings Analysis

Discussion – Current system comparison with previous process

If previous system is using manual process, add a table of user acceptance test to discuss the current system successfully minimize the manual process problems.



Result & Findings Analysis

Discussion – Current system comparison with previous process

DESCRIPTION	Previous System	Current System
Registration process	Manual paper form based usually takes 5 days to process 10-20 registration data	Takes less than a day to process 30 registration data in database
Update data	Require user to refill the paper form and admin need to replace the previous data form manually	Auto retrieve previous data and update from the database by the user

Add a paragraph to discuss the table:

Table 4.7 elaborates on the findings of the (put your system's name) developed. This (your system's name) tested by the (your user/client) and take less than a day to automatically process 30 registration data as compared to previous process where it took 5 days. In term of updating data .....

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Chapter Summary

ADD SUMMARY

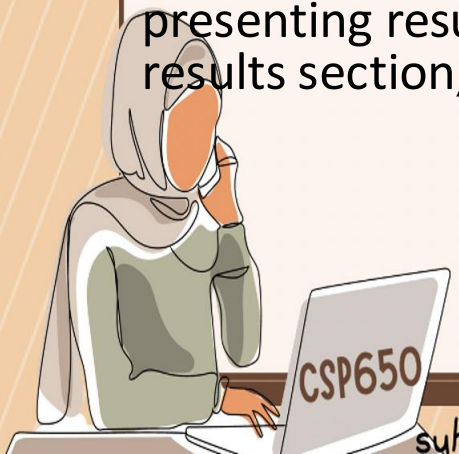
If you are putting your discussion into a discussion section separate from the results, you may want to **provide a summary** of the results to remind your reader of your main findings.

Put your results in **context** (e.g. by comparing them with previous project/research, or with existing theory) in order to explain them.

Give **reasons** to account for differences between your project/research and previous project/research or existing theory, or to explain unexpected results.

Although there may be some repetition of information in the results and discussion sections, it should be kept to a minimum.

Remember too that the focus should be different: while you are simply presenting results and making them meaningful to your reader in the results section, in the discussion section you are **explaining** them.



SUMMARY SAMPLE

4.5 Chapter Summary

This chapter discuss about result and finding that consist of all the deliverables for each phase of SDLC that this project undergoes. All results have been detailed and also explained how these results have reached the project objectives. In this chapter, there also have screenshots of the system interface, database architecture and also shows the implementation of data visualization towards this system.



THANK YOU !

See you in next Lecture!



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